

**B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
EXAMINATION, 2022**

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code : 510221

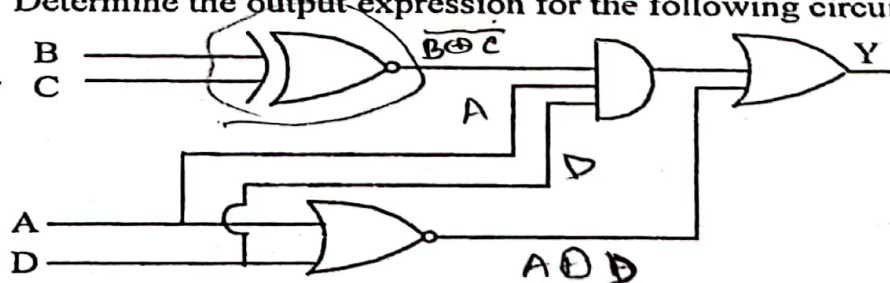
Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | | Marks |
|----|---|-------|
| 1. | (a) Describe the organization of a digital computer with figure. | 4 |
| | (b) What is binary number system? How any binary number can be converted into its equivalent decimal number and vice-versa? | 3 |
| | (c) Define the following terms with examples :
(i) Gray Code
(ii) BCD Code
(iii) Excess-3 Code
(iv) ASCII Code. | 8 |
| | (d) What do you mean by 2's complement? Using 2's complement system subtract 27 from 32. | 5 |
| 2. | (a) Implement XOR and XNOR gate using NAND gate. | 6 |
| | (b) Design a logic circuit that output is high only when a majority of inputs A, B and C are low. | 5 |
| | (c) State and prove the De-Morgan theorem with the help of truth table for three variables. | 5 |
| | (d) Determine the output expression for the following circuit : | 4 |



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		Marks
3.	(a) Write the algebraic function of exclusive-OR and exclusive-NOR gate? Also draw their symbol and truth table.	6
	(b) Using Boolean expression show that $F \cdot \bar{F} = 0$ if $F = \bar{x}y + xy\bar{z}$.	4
	(c) Describe the J-K flip-flop with internal diagram.	5
	(d) Minimize the following expression using k-map $F(A, B, C, D) = \sum m(1, 3, 6, 7, 11, 15) + d(0, 2, 4, 5)$.	5
4.	(a) What is sequential logic circuit? Differentiate between combinational circuit and sequential circuit.	5
	(b) What is Flip-Flop? What is NAND gate latch?	2+2=4
	(c) What is setup and hold times? Explain S-R Flip-Flop with suitable diagram.	6
	(d) What is race around condition? How it is resolved? Explain with example.	5
5.	(a) Design and construct full adder circuit. Explain its operation with truth table.	6
	(b) Write down the difference between synchronous and asynchronous counter.	4
	(c) Draw the logic diagram of 4-bit ring counter and explain its operation.	6
	(d) Design 8 : 1 multiplexer with truth table.	4
6.	(a) Write down the difference between RAM and ROM.	4
	(b) What is Decoder? Explain 3 to 8 line Decoder with truth table and circuit diagram.	6
	(c) Design a synchronous MOD—10 counter and show its timing diagram.	6
	(d) Simplify the expression $Z = \bar{A}C(\bar{A}BD) + \bar{A}BC\bar{D} + A\bar{B}C$ and draw the circuit diagram.	4

**B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
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DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

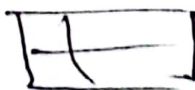
Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four of the following questions.]

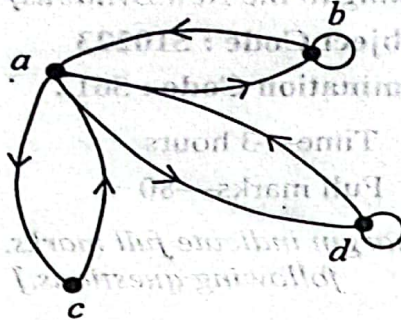
Marks

1. (a) Define proposition, conjunction, disjunction and implication with truth table. 4
- (b) Translate the following sentences into logical expressions : 2×3=6
 - (i) "You can take the flight if and only if you buy a ticket.
 - (ii) You can access the internet from campus only if you are a computer science major or you are not a freshman.
 - (iii) The automated reply cannot be sent when the file system is full.
- (c) Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. 5
- (d) Define Inclusion-Exclusion Principle. Suppose that $|A| = 5$, $|B| = 9$ and $|A \cap B| = 4$. Find $|A \cup B|$. 2+3=5
2. (a) Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers. 5
- (b) Define rules of inference. Write down the basic rules of inference. 4
- (c) Define one-to-one and onto correspondence functions with examples. 3
- (d) Express the statement "Every student in this class has studied calculus" as a universal quantification. 4
- (e) What is bit string? Find the bitwise OR, bitwise AND and bitwise XOR of the bit strings 011 011 0110 and 11 0001 1101. 4

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3. (a) What is relation? Write down the properties of relation. 5
 (b) Determine whether the relation for the following directed graph is reflexive, symmetric, antisymmetric and transitive : 5



- (c) Consider the following relations on the set of integers :

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

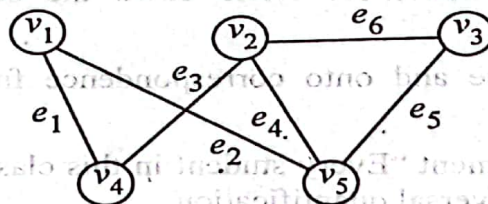
$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contains each of the pairs (1, 1), (1, 2), (2, 1), (1, -1) and (2, 2)?

- (d) Give an indirect proof of the theorem "if $3n + 2$ is odd, then n is odd". 4

4. (a) What is Graph? How many edges are there in a graph with 10 vertices each of degree six? 4

- (b) Define Adjacency and Incidence matrix. Represent the graph shown in figure with an incidence matrix. 5



- (c) Define multigraph and pseudograph with example. 3

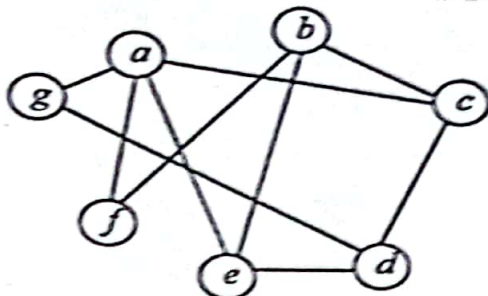
- (d) Draw the directed graph for the relation : 3

$$R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$$

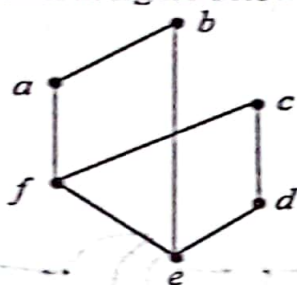


a) c e f g

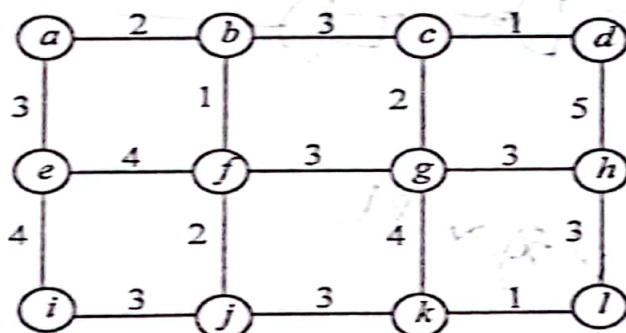
- (e) Define bipartite graph. Is the following graph bipartite?



5. (a) Write down the Handshaking theorem. How many edges are there in a graph with 10 vertices each of degree 6? $2+3=5$
~~(b)~~ State Euler's Formula and prove it. 5
 (c) Define planar graph with example. Is $K_{3,3}$ (complete bipartite graph) planar graph or not. $2+3=5$
 (d) Is the graph shown in figure below bipartite? Justify your answer. 5



6. (a) What is minimum spanning tree? Use Prim's algorithm to find a minimum spanning tree in the graph shown in the following figure: 5



- (b) Construct a Full Adder circuit using Half Adder circuits. 5
 (c) Simplify the sum-of-products expansions using Karnaugh map: $3 \times 2 = 6$
 (i) $xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$
 (ii) $xyz + xy\bar{z} + x\bar{y}z + \bar{x}yz$
 (d) Define Cartesian product. What is the Cartesian product $2 \times 2 = 4$ $A \times B$, where $A = \{0, 1\}$ and $B = \{a, b, c\}$?

**B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
EXAMINATION, 2022**

LINEAR ALGEBRA

[According to the New Syllabus]

Subject Code : 510225

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) What do you mean by system of linear equation. Solve the following system of linear equation : 2+6=8

$$x_1 + 2x_2 - x_3 + 3x_4 = 3$$

$$2x_1 + 4x_2 + 4x_3 + 3x_4 = 9$$

$$3x_1 + 6x_2 - x_3 + 8x_4 = 10.$$

- (b) Solve the following system of linear equations : 6

$$3x - y + z = -2$$

$$x + 5y + 2z = 6$$

$$2x + 3y + z = 0.$$

(c) If $\Delta = \begin{vmatrix} x & x^2 & x^3 + 1 \\ y & y^2 & y^3 + 1 \\ z & z^2 & z^3 + 1 \end{vmatrix}$, then prove that, 6

$\Delta = (x - y)(y - z)(z - x)(xyz + 1)$. Also show that if x, y and z are not equal and $\Delta = 0$, then $xyz + 1 = 0$.

2. (a) Define : Singular Matrix, Nilpotent Matrix and Hermitian Matrix. 2+3=6

- (b) If A and B are non-singular matrices, then prove that $(AB)^{-1} = B^{-1}A^{-1}$. Also prove that $(A^{-1})^{-1} = A$ and $(A^{-1})^t = (A^t)^{-1}$. 4+1+1=6

- (c) Define inverse of matrix. Find inverse of matrix, 2+6=8

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix}.$$

[Please turn over]

Marks

3. (a) If $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ then prove that, $A^3 + A^2 - 21A - 45I = 0$ 8

where I is the identity matrix of order 3×3 .

- (b) Define orthogonal matrix. And show that the matrix, 2+5=7

$A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is idempotent.

- (c) Solve the following system of linear equation by Matrix method : 5

$x_1 + 3x_2 + 2x_3 = 5$

$2x_1 + x_2 + 3x_3 = 1$

$3x_1 + 2x_2 + x_3 = 4$

4. (a) Define norm of a vector in $\mathbb{R}^n$. Consider $A(3, \lambda, -2)$ and $B(5, 3, 4)$ in $\mathbb{R}^3$. Find the value of λ so that, $\overrightarrow{AB}$ is orthogonal to the vector $(4, -3, 2)$. 6

- (b) If $\overline{u} = (2 - 5i, 3i, -1 + i)$, $\overline{v} = (4 - i, 8 + 2i, 7 - 3i)$ then find $\overline{u} \cdot \overline{v}$, $\|u - v\|$. 6

- (c) What is unit vector in $\mathbb{R}^n$? For any vectors $u, v \in \mathbb{R}^n$, then prove that, $\|u \cdot v\| \leq \|u\| \cdot \|v\|$. 2+6=8

5. (a) Define linear transformation. Let U and V be two vector spaces over the field F and T_1 and T_2 be linear transformations from U into V then prove that, $T_1 + T_2$ is a linear transformation. 2+5=7

- (b) If T_1 and T_2 be the linear transformation $T_1, T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T_1(x, y, z) = (x + y - z, 2y + z)$ and $T_2(x, y, z) = (y, z)$, then find (i) $(T_1 T_2)(1, 0, 1)$, (ii) $(T_2 T_1)(1, 0, 1)$, (iii) $(T_1 + T_2)(1, 0, 1)$, (iv) $(T_1 - T_2)(1, 0, 1)$. 8

- (c) Show that the mapping $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (3x + 2y - 4z, x - 5y + 3z)$ is linear. 5

6. (a) Define eigen values, eigen vectors and characteristic Matrix. 4

- (b) State and prove Cayley-Hamilton theorem. 6

- (c) Find the eigen values and the corresponding eigen vectors of 10

the matrix, $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$.

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**B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
EXAMINATION, 2022**

STATISTICS AND PROBABILITY

[According to the New Syllabus]

Subject Code : 510227

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|--|-------|
| 1. (a) What is data? Write different sources of statistical data. | 4 |
| (b) Describe the following random sample of <u>10</u> final exam grades for a class with a stem and leaf display :
88, 51, 63, 85, 79, 65, 79, 70, 73, 77. | 6 |
| (c) The following is the number of commute from home to work for a group of automobile executives :
28, 25, 48, 37, 41, 19, 32, 26, 16, 23, 23, 29
36, 31, 26, 21, 32, 25, 31, 43, 35, 42, 38, 33
(i) Construct a frequency distribution table with suitable class interval.
(ii) Draw a relative frequency polygon. | 10 |
| 2. (a) Define measures of central tendency. Which measure of central tendency is best and why? | 5 |
| (b) Prove that, $\log G = \frac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$ | 5 |
| (c) The daily profits in Taka of 100 shops are distributed as follows : | 10 |

Profits	0—100	100—200	200—300	300—400	400—500
No. of shops	12	18	27	20	17

Calculate mean, median and mode by given data.

[Please turn over

3. (a) Distinguish between absolute and relative measures of dispersion. 3
- (b) Define Kurtosis. Explain different types of Kurtosis with the help of diagram. 5
- (c) Prove that, for two unequal observations, $MD = SD = R/2$, where, MD = Mean deviation about mean, SD = standard deviation and R = Range. 5
- (d) The demand of a commodity in two different cities are given below : 7

City-A	40	44	18	36	46	60
City-B	42	27	26	43	32	40

In which city the demand condition is more stable and why?

4. (a) When a correlation is called Simple and Multiple? 4
- (b) Show that the limit of correlation coefficient is -1 to $+1$. 4
- (c) When use rank correlation? Compute the rank correlation coefficient for the following data of the marks obtained 8 student in statistics and programming language : 8

Marks in statistics	15	20	30	55	20	70	20	65
Marks in prog. language	40	25	45	65	30	40	65	78

- (d) What is scatter diagram method? With scatter diagram show that $r = +1$, $r = 0$, $r = -1$. 4
5. (a) What is regression equation? How you calculate regression parameters for multiple regression model. 4
- (b) The following information given below : 8

Advertising expenditure (X)	5	4	3	4	5
Sales (Y)	10	8	3	8	7

- (i) Find regression equation Y on X.
- (ii) Plot the regression line on the graph paper.

- (c) Given the following values of X_1 , X_2 and X_3 . Compute the multiple correlation coefficient $R_{1.23}$.

Marks
8

X_1	5	8	4	6	3
X_2	4	2	3	7	5
X_3	3	5	1	8	4

6. (a) Define association of attributes. Define different types of association. 5
- (b) The male population of a certain state is 250 lakhs. The number of literate males is 26 lakhs and the total numbers of male criminals is 32 thousand. The number of literate male criminal is 3,000. Do you find any association between literary and criminality? 5
- (c) Calculate spearman's rank-correlation from the given data : 5

x	40	62	55	62	58	40	45
y	44	44	44	50	32	25	37

- (d) Calculate co-efficient of skewness by following data : 5

Marks	30—40	40—50	50—60	60—70	70—80
No. of students	16	24	30	40	20

**B.Sc (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2022**

HISTORY OF THE EMERGENCE OF INDEPENDENT BANGLADESH

Subject Code—510229

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions. Sequence must be maintained in answering each of the questions.]

- | | Marks |
|--|-------|
| 1. (a) Briefly discuss the emergence of Independent Bangladesh. | 7 |
| (b) Discuss the background of Lahore Resolution. | 6 |
| (c) Write a short note on the partition of Bengal. | 3 |
| (d) Discuss the importance of Language movement. | 4 |
| [(ক) স্বাধীন বাংলাদেশের অভ্যুদয়ের ইতিহাস সংক্ষেপে বর্ণনা কর। | |
| (খ) ঐতিহাসিক লাহোর প্রস্তাবের প্রেক্ষাপট তুলে ধর। | |
| (গ) বঙ্গভঙ্গ সম্পর্কে টীকা লেখ। | |
| (ঘ) ভাষা আন্দোলনের গুরুত্ব আলোচনা কর।] | |
| 2. (a) Discuss the working of Basic Democracy practice. | 4 |
| (b) What were the programs of 11 Points movements. | 4 |
| (c) Describe the causes of the mass upsurge of 1969. | 6 |
| (d) Discuss the result of election of 1970. | 6 |
| [(ক) মৌলিক গণতন্ত্রের কার্যকারিতা আলোচনা কর। | |
| (খ) ছাত্রদের ১১ দফা আন্দোলনের কর্মসূচী কী ছিল? | |
| (গ) ১৯৬৯ সালের গণঅভ্যুত্থানের কারণসমূহ আলোচনা কর। | |
| (ঘ) ১৯৭০ সালের নির্বাচনের ফলাফল আলোচনা কর।] | |
| 3. (a) Discuss the challenges of Bangabandhu Government. | 7 |
| (b) Describe the features of the constitution of 1972. | 6 |
| (c) Discuss the causes of success of the united front in the election of 1954. | 5 |
| (d) What do you mean by operation search light? | 2 |
| [(ক) বঙ্গবন্ধু সরকারের চ্যালেঞ্জসমূহ আলোচনা কর। | |
| (খ) ১৯৭২ সালের সংবিধানের বৈশিষ্ট্যসমূহ বর্ণনা কর। | |
| (গ) ১৯৫৪ সালের নির্বাচনে যুক্তফ্রন্টের বিজয়ের কারণসমূহ আলোচনা কর। | |
| (ঘ) অপারেশন সার্চ লাইট বলতে কী বোঝ? | |

[Please turn over

		Marks
4.	(a) Explain the background of establishment of the Awami League.	4
	(b) What is United Front? Discuss the background and results of United front Election.	6
	(c) Economic deprivation during Pakistan Period is responsible for the development of Bangladesh.—Explain.	6
	(d) What is the main four principles of the constitution of Bangladesh? Write in brief.	4
	[(ক) আওয়ামী লীগ প্রতিষ্ঠার পটভূমি ব্যাখ্যা কর।	
	(খ) যুক্তফ্রন্ট কী? যুক্তফ্রন্ট নির্বাচনের পটভূমি ও ফলাফল আলোচনা কর।	
	(গ) বাংলাদেশের অভ্যুদয়ে পাকিস্তান আমলের অর্থনৈতিক বঞ্চনাই দায়ী—ব্যাখ্যা কর।	
	(ঘ) বাংলাদেশের সংবিধানের মূল চারনীতি কী? সংক্ষেপে লেখ।]	
5.	(a) Why six points Program is called the “Magna Carta” of the Bangalee?	6
	(b) What were the causes and results of the Agartala conspiracy case?	4
	(c) Critically discuss the role of U.S.A and U.S.S.R in the liberation war of Bangladesh.	6
	(d) Discuss Martyr’s Day and International Mother Language Day.]	4
	[(ক) ছয় দফা দাবিকে কেন বাঙালীর “ম্যাগনা কার্টা” বলা হয়?	
	(খ) আগরতলা মামলার কারণ কী ছিল? এর পরিণতি কী হয়েছিল?	
	(গ) বাংলাদেশের মুক্তিযুদ্ধে আমেরিকা এবং সোভিয়েত ইউনিয়ন (বর্তমান রাশিয়া) এর ভূমিকা আলোচনা কর।	
	(ঘ) শহীদ দিবস এবং আন্তর্জাতিক মাতৃভাষা দিবস সম্পর্কে আলোচনা কর।]	
6.	(a) Write a short note on the Mujibnagar Government.	4
	(b) Write down the importance of the 7 th , March address of Bangabandhu.	4
	(c) Give the ethnic identity of the people of Bangladesh.	7
	(d) Discuss the ‘two nations’ theory.	5
	[(ক) মুজিবনগর সরকার সম্পর্কে টীকা লেখ।	
	(খ) ৭ মার্চ বঙ্গবন্ধুর ভাষণের গুরুত্ব লেখ।	
	(গ) বাংলাদেশের জনগণের নৃ-তাত্ত্বিক পরিচয় দাও।	
	(ঘ) ‘দ্বি-জাতি’ তত্ত্ব সম্পর্কে আলোচনা কর।]	